Free Choice for Weak Permission: A conditional analysis

Dean McHugh*

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Abstract

According to an influential approach to deontic modals, permission is conditional and strong. On this view, to say that Alice is permitted to eat an apple is to say something along the lines of: if Alice ate an apple, the relevant deontic ideals would be met. This paper argues against the strong analysis. We present a variety of counterexamples to the strong conditional analyses of permission, which instead support the weak analysis that something is permitted just in case, were it true, there would be some case where the deontic ideals are met.

With this change, however, we lose a central motivation for the conditional analysis of permission: its promising account of free choice. The strong conditional analysis reduces free choice to Simplification of Disjunctive Antecedents (SDA). This paper shows how to preserve the reduction of free choice to SDA even for weak permission, via alternative semantics. What emerges is a new theory of possibility on which there is a systematic connection between modals and conditionals.

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1 Introduction

According to an influential approach to deontic modals, permission is *conditional* and *strong* (von Wright 1968, Hilpinen 1982, Lokhorst 1997, Asher and Bonevac 2005, Barker 2010, Li 2025). On this view, to say that Alice may eat an apple (or has permission to, or is allowed to, and so on) is to say something along the lines of: If Alice ate an apple, the relevant deontic ideals would be met; things would be deontically 'OK'. It is *conditional* in the sense of borrowing components from the theory of conditionals in the analysis of permission. And it is *strong* in the sense that '*A* is permitted' expresses the conditional 'if *A*, OK' rather than its dual, 'it is not the case that: if *A*, not OK'. For *A* to be permitted on the strong analysis, supposing *A* must *guarantee* that things are OK, while on the weak analysis, it need only be *compatible* with things being OK.

The overwhelming majority view among conditional analyses of permission today is that permission is strong. Against this, I argue for the weak analysis. My argument comes from mixed cases, where supposing 'if *A*' results in some cases where the deontic ideals are met and some where they are violated. In such cases, we observe that permission statements pattern with the weak conditional, rather than the strong conditional. I conclude that if one wishes to maintain the conditional analysis of permission, the conditional involved should be weak: *A* is permitted iff supposing 'if *A*' results in *some* case where the deontic ideals are met.

With this change, however, comes an unfortunate consequence. We lose a central motivation for the conditional analysis of permission: its promising account of free choice. As is well known (Kamp 1973, Merin 1992), '*A* or *B* is permitted' intuitively implies both '*A* is permitted' and '*B* is permitted', an inference pattern known as *free choice*. The conditional analysis of permission accounts for free choice by reducing it to a popular principle of conditional logic, that of Simplification of Disjunctive Antecedents (SDA): the inference from 'if *A* or *B*, *C*' to 'if *A*, *C*' and 'if *B*, *C*'. On the conditional analysis, 'Alice may eat an apple or a pear' states that if she ate an apple or a pear, things would be OK. SDA then implies that if she ate an apple, things would be OK, and if she ate a pear, things would be OK. On the conditional theory, this is equivalent to: Alice may eat an apple and she may eat a pear—the desired result.

This account requires permission to be strong to guarantee SDA. For on many analyses of the conditional that have been plugged into the conditional analysis of permission, the strong form validates SDA but its weak dual does not (Asher and Bonevac 2005:306). This holds for the strict conditional (von Wright 1968, Li 2025), the variably strict conditional (Hilpinen 1982, Nute 1985, assuming that there are *A*-worlds and *B*-worlds among the closest *A or B*-worlds; see Lewis 1977), the relevance conditional (McArthur 1981, Mares 1992), the conditional of non-monotonic logic (Asher and Bonevac 2005), and the conditional of linear logic (Barker 2010). Each only validates SDA for the strong conditional.

However, I show how a weak conditional analysis of permission can recover the reduction of free choice to SDA by using alternatives. A number of researchers have recently argued that conditional antecedents are interpreted via sets of propositions, called their 'alternatives', and that a conditional is true just in case, for each alternative, supposing it results in the truth of the consequent (Alonso-Ovalle 2006, 2009, Ciardelli 2016, Santorio 2018, Willer 2018, Cariani and Goldstein 2020, Khoo 2022). The conditional reduction of modality offers an automatic way to extend this apparatus to modals. The resulting theory reduces free choice to SDA In this way, alternatives allow us to have our cake and eat it: to reduce free choice to SDA even for weak permission.

My goal is to argue that if we want a conditional analysis of permission, it should be weak rather than strong. I therefore take as a working assumption that permission has some conditional analysis or other. There are enough conditional analyses of permission available to make this a worthwhile exercise (Anderson 1956, 1967, von Wright 1968, Hilpinen 1982, Lokhorst 1997, Asher and Bonevac 2005, Barker 2010, Li 2025). Beyond conditional analyses, the present study also serves as an interesting test case for the broader agenda of providing a uniform account of free choice and SDA; pursued, for example, by van Rooij (2006), Klinedinst (2007), Franke (2011), Willer (2018), and Bar-Lev and Fox (2020). Motivation for a uniform account comes not only by the parallel behaviour of modals and conditional antecedents discussed in this paper, but also by recent acquisition data (Zani, Ciardelli, and Sanfelici 2026). In addition, the present account bears on the extensive literature discussing permission in Japanese, Korean, and Burmese, languages that typically express permission using conditionals (see Anderson 1956, Clancy 1985, Akatsuka 1992, Wymann 1996, Nauze 2008, Knoob 2008, Narrog 2009, Kaufmann 2017a, Chung 2019, among others).

The paper proceeds as follows. Section 2 outlines the conditional reduction of deontic logic. Section 3 presents the challenge from mixed cases and shows how they differ from previous challenges to the strong conditional theory. Section 4 shows how weak permission without alternatives runs into the problem of free choice permission. Section 5 surveys Alonso-Ovalle's treament of conditionals within alternative semantics and shows how it automatically extends to modals via the conditional reduction. Section 6 argues for an ambiguity approach to SDA and shows how this generates novel, intuitively correct predictions for permission statements under probability operators. Section 7 develops some extensions of the basic template, generalising the account to arbitrary modal flavours (including epistemic and ability modals), providing a first-order variant, and accounting for deontic gaps (something Barker 2010 took to motivate a strong theory of permission). Section 8 compares our account with existing alternatives. Section 9 concludes.

2 The conditional reduction of deontic logic

We start with a simple formalisation of the conditional theory of permission. The theory can be refined in myriad ways, but will serve as a helpful starting point to fix ideas. We assume a standard propositional language $\mathscr{L}$ built up from countably many atomic sentences $At = \{\mathrm{p}, \mathrm{q}, \dots\}$, negation ($\neg$), conjunction ($\wedge$), and disjunction ($\vee$). In addition, we introduce two unary modal operators $\Box$ and $\Diamond$, and conditional operators $\Box\!\!\rightarrow$ and $\Diamond\!\!\rightarrow$.[1]

$$A ::= \mathrm{p} \mid \neg A \mid A \wedge B \mid A \vee B \mid \Box A \mid \Diamond A \mid A \mathrel{\Box\!\!\rightarrow} B \mid A \mathrel{\Diamond\!\!\rightarrow} B$$

We define a *model* M to be a tuple (W, f, V, OK), where

- W is a non-empty set (the set of possible worlds);
- $f : \wp(W) \times W \to \wp(W)$ is a conditional selection function taking a proposition p (a set of worlds) and a world, and returning a subset of p (those that result from supposing the proposition true at that world);
- $V : At \to \wp(W)$ is a valuation function, taking an atomic sentence and returning a set of worlds (those at which it is true).
- $\mathrm{OK} : W \to \wp(W)$ is a function taking each world and returning a set of worlds (where the deontic ideals at that world are met).

2.1 Semantic clauses

Given a model M, we define an interpretation function $[\![\cdot]\!]_M$ determining the set of worlds at which each sentence is true as follows (we omit the subscript M when clear from context; for any set of worlds P, let $\overline{P}$ denote P's complement, $W \setminus P$).

$$\begin{aligned}
[\![\mathrm{p}]\!] &= V(\mathrm{p}) & [\![\Box A]\!] &= \{w : f([\![\neg A]\!], w) \subseteq \overline{\mathrm{OK}_w}\} \\
[\![\neg A]\!] &= \overline{[\![A]\!]} & [\![A \mathrel{\Diamond\!\!\rightarrow} C]\!] &= \{w : f([\![A]\!], w) \cap [\![C]\!] \neq \emptyset\} \\
[\![A \wedge B]\!] &= [\![A]\!] \cap [\![B]\!] & [\![A \mathrel{\Box\!\!\rightarrow} C]\!] &= \{w : f([\![A]\!], w) \subseteq [\![C]\!]\} \\
[\![A \vee B]\!] &= [\![A]\!] \cup [\![B]\!]
\end{aligned}$$

The clauses for the connectives are classical. Something is a deontic necessity just in case, if it were false, the rules would be broken. As Anderson (1967:346–47) puts it, 'it is *analytic* of the notion of obligation that if an obligation is not fulfilled, then something has gone wrong'. On this view, 'You must not murder' means that if you murder, necessarily, the rules have been broken. The universal conditional $A \mathrel{\Box\!\!\rightarrow} C$

[1] Note that we take *statements* to be permitted, in line with the standard theory of permission (Prior 1955, Hanson 1965, Kratzer 1977) and the conditional reduction (Anderson 1956, Kanger 1957). One may alternatively take *action types* to be permitted, as von Wright (1951) and Castañeda (1975) do, without affecting the arguments to come.

holds iff every selected A-world is a C-world, while the existential conditional $A \diamond\!\!\rightarrow C$ holds iff some selected A-world is a C-world.

The strong and weak analyses of possibility are:

$$\begin{aligned} \textit{Strong}: &\quad [\![\diamond A]\!] = \{w : f([\![A]\!], w) \subseteq \mathrm{OK}_w\} \\ \textit{Weak}: &\quad [\![\diamond A]\!] = \{w : f([\![A]\!], w) \cap \mathrm{OK}_w \neq \emptyset\} \end{aligned}$$

On the strong theory, A is permitted iff *every* selected A-world is OK; on the weak theory, A is permitted iff *some* selected A-world is OK.

Some remarks on the formalisation:

The selection function. Our formalisation of the conditional follows Chellas (1974:29, 1980:268), who takes conditional selection functions to express the general semantics of a conditional (Section 8.1 explores different interpretations of the conditional, from relevance, non-monotonic, and linear logic). Different theories will interpret the selection in different ways, and thereby impose different constraints on it. For instance, on variably strict theories (Stalnaker 1968, Lewis 1973) given the limit assumption (see Kaufmann 2017b), $f(p,w)$ will contain the closest p-worlds to w; on strict theories (von Fintel 2001a, Gillies 2007), $f(p,w)$ will contain the p-worlds in w's modal horizon.

If we take $\Box\!\!\rightarrow$ to express the conditional in natural language, one question that arises is whether it expresses the indicative or subjunctive (also known as counterfactual, past tense, or X-marked conditional). There is evidence that, between the indicative and subjunctive, the subjunctive gives better predictions for permission when plugged into the conditional reduction. A security guard who is required to stay at their post might well say, 'I am required to be at my post right now' or 'I'm not allowed to be elsewhere right now'. On the conditional reduction using the subjunctive, these express 'If I were not at my post right now, the rules would be broken'—a perfectly fine meaning. On the indicative, they express 'If I am not at my post right now, the rules are/will be broken', which is certainly odd.[2] This is expected: the interpretation of indicatives, unlike subjunctives, is commonly thought to be restricted to worlds compatible with the common ground (Stalnaker 1975). The guard knows that they are at their post, but there is no hint of oddness in them saying that they are required to be there, or not allowed to be elsewhere right now.

The notion of OK. Our formalisation provides an especially simple model of OK—what it means for the deontic ideals to be met: each world w provides a binary classification of worlds into those where the deontic ideals are met (the OK_w worlds) and those where they are violated (the not OK_w worlds; for simplicity we often omit the subscript

[2]The 'right now' ensures that the present tense is not interpreted generically, as in 'If I am not at my post, the rules are/will be broken', which is acceptable even when the guard is at their post.

w when clear from context). This picture is compatible with Kratzer's (1977) influential analysis of modality, whereby we take the OK worlds at w to be the 'best' worlds at w; that is, the worlds in the modal base at w coming closest to satisfying the relevant ideals at w.[3] That being said, the precise analysis of 'OK' will not affect our points to come, so for simplicity, we do not commit to one particular analysis of 'OK' here.

3 Mixed cases

According to the strong conditional analysis of permission, '*A* is permitted' is equivalent to a conditional stating that if *A* were true, the relevant deontic ideals would be met. The following cases put this idea to the test.

(1) *There are two switches, A and B. Each switch can be either up or down. The rules specify that A must be up and that B may be in any position.*

   a. Are the positions of the switches allowed to agree?
   b. If the positions of the switches agreed, would the rules be met?

(2) *Angelica has many allergies. She must not eat soy, gluten, fish, or nuts.*

   a. Is Angelica allowed to eat food?
   b. If Angelica ate food, would her dietary rules be met?

(3) *Alice and Bob work together in Quebec. They both know English and French, and talk to each other in English. A new rule requires all employees to speak French at work. Still, even after the rule is introduced, Alice and Bob continue to communicate in English (even though they are able to speak French).*

   a. Does the new rule permit Alice and Bob to talk to each other at work?
   b. If Alice and Bob talked to each other at work, would the new rule be met?

(4) *John's doctor has permitted him to smoke at most seven cigarettes per day.*

   a. Is John allowed to smoke more than six cigarettes per day?
   b. If John smoked more than six cigarettes per day, would his doctor's rules be met?[4]

In each pair we observe a strong contrast between the permission statement and the corresponding conditional. The intuitive answer to each permission question is 'Yes', while we are much more inclined to respond to each conditional question with 'Indeterminate' or 'It depends'.

[3] Formally, context supplies $b, g : W \to \wp(\wp(W))$, called the *modal base* and *ordering source*, respectively. We define: $w' \leq_{g(w)} w''$ iff for all $p \in g(w)$, $w'' \in p$ implies $w' \in p$; $w' <_{g(w)} w''$ iff $w' \leq_{g(w)} w''$ but not $w'' \leq_{g(w)} w'$; and $\textsc{best}(b,g,w) = \{w' \in \bigcap b(w) : \neg\exists w'' \in \bigcap b(w) : w'' <_{g(w)} w'\}$. Given b and g, we take $\text{OK}_w = \textsc{best}(b,g,w)$.

[4] This example comes from Fox (2007b:103).

There are also naturally occurring examples where permission is weak:

> Am I allowed to have help when I am voting? Yes! In many forms.
>
> (Maine Students Vote 2025)

> The injured party is allowed to be present during investigative acts in the phase of a judicial investigation. Specifically, he is allowed to be present at [...] the questioning of a witness, but only if it is expected that the witness will not come to the trial.
>
> (Zgaga 2011:89)

You are allowed to have help when voting in many forms, but presumably not all. Being present during investigative acts is allowed, but not in all cases. Compare these with the corresponding conditionals:

(5) a. If I have help when I am voting, the rules will be met.
b. If the injured party is present during investigative acts in the phase of a judicial investigation, the rules will be met.

While the permission statements can be asserted outright, asserting the corresponding conditionals in each context does not seem appropriate.

Summing up, mixed cases pose a serious challenge to the strong analysis of permission, on which the permission statement and conditional are taken to be equivalent. Section 8 shows in detail how existing strong analyses of permission make incorrect predictions for these cases. They favour the weak analysis, on which A is permitted iff *some* selected A-world is OK.

3.1 Strong inferences

Before proceeding, it is worth addressing one complication that may arise when discussing mixed cases. The complication is that, while the answers to the permission questions in (1)–(4) are intuitively 'Yes', there is nonetheless something misleading about asserting some of the permission statements outright in their respective contexts:

(6) a. The switches are allowed to agree.
b. Alice and Bob are allowed to talk to each other at work.
c. John has permission to smoke more than six cigarettes per day.

This is especially clear if these statements are not given in response to a Yes/No question, asking whether or not the statement is true, but in response to an open-ended question about what is permitted. In the case of (6a), for example, if one is asked 'What positions of the switches are allowed?' or 'Tell me about the rules governing the switches', the answer 'The switches are allowed to be in the same position' seems deeply misleading. It naturally suggests that *every* selected world where the switches

are in the same position is allowed; that they may both be up and that they may both be down. The same goes for the other examples in (6), which in response to such questions have the inferences in (7), respectively.

(7) a. The switches are allowed to be up and they are allowed to be down.
b. Alice and Bob are allowed to talk to each other at work in any language.
c. John has permission to smoke seven, eight, nine, ... cigarettes per day.

If permission is strong, these inferences fall out as entailments. One might therefore be tempted to conclude that permission is strong after all. However, I believe they are best captured as pragmatic strengthenings of a weak literal meaning. They have the hallmarks of pragmatic inference. First, they are highly context-sensitive. As we have just seen, they are sensitive to the question under discussion. In response to the polar question 'Is *A* permitted?', the natural answer to the permission questions (1a)–(4a) is 'Yes', while in response to a more open-ended question such as 'Under what conditions is so-and-so permitted?' or 'Tell me about the rules governing so-and-so', responding with '*A* is permitted' is often unacceptable. Such sensitivity to the question under discussion is a characteristic sign of a pragmatic inference (Welker 1994, von Kuppevelt 1996, Breheny, Katsos, and Williams 2006, Zondervan 2009, Degen and Tanenhaus 2015, Ronai and Xiang 2021).

Second, they are cancellable. It is consistent to say, 'The switches are allowed to be in the same position, but they are not allowed to be both down'. In contrast, 'Any configuration where the switches are in the same position is allowed, but they are not allowed to both be down' is contradictory.

Third, denying the inference has the telltale prosodic contour of denying an implicature, often called metalinguistic negation. There is a distinctive prosodic contour in, for example, 'Max didn't talk to Anna OR Robin, he talked to BOTH', 'Arseny doesn't LIKE mushrooms, he LOVES them', and, from Horn (1985), 'I didn't trap two monGEESE, I trapped two monGOOSES'. The exact same prosodic contour appears in, say, 'Alice is not allowed to paint the house RED, she is allowed to paint it SCARLET'.

4 Free choice

Mixed cases provide strong evidence that permission is weak. However, weak permission runs straight into the problem of free choice permission (Kamp 1973, Merin 1992). Prior noticed a version of this problem back in 1959:

> [W]e would not ordinarily infer "It is permitted either to smoke or to embezzle" from "It is permitted to smoke." [...] but α (say my feeding this man) is usually considered permissible even if some ways of doing α (e.g. feeding him with poison) are not, so long as some are; and smoking is one

way of smoking-or-embezzling.

(Prior 1959:178)

Prior identifies an important tension. On the one hand, permission seems to have a weak, existential semantics. On the other, 'A is permitted' does not imply 'A or B is permitted'. These pull us in opposite directions.

While Prior phrases his point in terms of 'ways of doing' an action, his tension remains if we rephrase it in terms more familiar to conditional theories of permission. When we suppose 'if you smoked or embezzled', plausibly, in at least one of the worlds we consider, you smoke and do not embezzle. This world is OK (smoking is—at least was by 1950s standards—OK). Since some selected smoking-or-embezzling world is OK, the weak analysis of permission undesirably predicts 'You may smoke or embezzle' to be true.

The problem is fully general. Whenever some selected $A \vee B$-world is OK, the weak analysis incorrectly predicts $\Diamond(A \vee B)$ to be true, even if no B-world is OK.

The strong analysis avoids this problematic prediction: to be permitted to smoke or embezzle, *all* selected worlds where one does so must be OK, including embezzling-worlds. As we saw above, the strong analysis has a plausible story to tell about free choice, the inference from $\Diamond(A \vee B)$ to $\Diamond A \wedge \Diamond B$, reducing it to SDA, the inference from $(A \vee B) \mathrel{\Box\!\!\rightarrow} C$ to $(A \mathrel{\Box\!\!\rightarrow} C) \wedge (B \mathrel{\Box\!\!\rightarrow} C)$. The weak analysis instead reduces free choice to SDA's existential analogue: the inference from $(A \vee B) \mathrel{\Diamond\!\!\rightarrow} C$ to $(A \mathrel{\Diamond\!\!\rightarrow} C) \wedge (B \mathrel{\Diamond\!\!\rightarrow} C)$.

4.1 Problems with the strong theory

In light of the strong theory's account of free choice, one might prefer to keep it and instead seek an account of the discrepancy between permission statements and conditionals in mixed cases. However, the strong theory's reduction of free choice to SDA has problems of its own. The overarching issue is that disjunction is 'special' in the following sense: two necessarily equivalent sentences can behave differently under permission, where the only relevant difference between them is that one is disjunctive and the other is not.

To illustrate, suppose a concert venue limits the number of tickets each customer may buy to at most five. Is Alice allowed to buy more than four tickets? Intuitively, she is, since she may buy five. Is she allowed to buy five or more tickets? Now we want to say that she is not, since she may not buy more than five tickets.[5]

(8) Alice is allowed to buy more than four tickets.

a. $\not\rightsquigarrow$ Alice is allowed to buy five tickets.

b. $\not\rightsquigarrow$ Alice is allowed to buy more than five tickets.

[5] '$\rightsquigarrow$' denotes an intuitive felt inference, while remaining neutral on its precise status.

(9) Alice is allowed to buy five or more tickets.

a. $\rightsquigarrow$ Alice is allowed to buy five tickets.

b. $\rightsquigarrow$ Alice is allowed to buy more than five tickets.

But Alice buys more than four tickets if and only if she buys five or more tickets. The two sentences are necessarily equivalent. This is a problem for strong theories of permission whose conditional validates Substitution (also known as LLE and RCEA): the rule that whenever A and B are necessarily equivalent—true in all the same worlds—$A \mathrel{\Box\!\!\rightarrow} C$ and $B \mathrel{\Box\!\!\rightarrow} C$ have the same truth value. A wide range of theories of the conditional that have been used to give a conditional analysis of permission validate Substitution. These include the strict conditional (Anderson 1956, von Wright 1968, Li 2025), the variably strict conditional à la Stalnaker and Lewis (Hilpinen 1982), and the conditional of non-monotonic logic (Asher and Bonevac 2005). Taking C to be true at all and only the OK-worlds, Substitution with strong permission predicts that whenever A and B are necessarily equivalent, A is permitted iff B is.

A similar problem arises for theories that aim for a uniform account of SDA and free choice, such as Willer (2018) and Li (2025), that preserve both De Morgan's law (the equivalence of $\neg(A \wedge B)$ and $\neg A \vee \neg B$) and the duality of permission and obligation (the equivalence of $\neg\Box A$ and $\Diamond\neg A$). Consider (10).

(10) a. Out of *Beloved* and *Song of Solomon*, you are required to read *Beloved*, but you are not required to read both of them. $\Box A \wedge \neg\Box(A \wedge B)$

b. Out of *Beloved* and *Song of Solomon*, you are required to read *Beloved*, but you are permitted to skip either of them. $\Box A \wedge \Diamond(\neg A \vee \neg B)$

The first is fine, while the second is odd: its second conjunct seems to contradict the first.[6] By De Morgan's law, $\neg(A \wedge B)$ is necessarily equivalent to $(\neg A \vee \neg B)$. Then assuming duality of $\Box$ and $\Diamond$ and Substitution, $\neg\Box(A \wedge B)$ is equivalent to $\Diamond(\neg A \vee \neg B)$. Intuitively, however, they are not equivalent:

(11) You are not required to read both A and B. $\neg\Box(A \wedge B)$

a. $\not\rightsquigarrow$ You are not required to read A. $\neg\Box A$

b. $\not\rightsquigarrow$ You are not required to read B. $\neg\Box B$

(12) You are permitted to skip either A or B. $\Diamond(\neg A \vee \neg B)$

a. $\rightsquigarrow$ You are permitted to skip A. $\Diamond\neg A$

b. $\rightsquigarrow$ You are permitted to skip B. $\Diamond\neg B$

[6] Ciardelli, Zhang, and Champollion (2018) make the same point for epistemic modals, observing that 'Mary might not speak both Arabic and Bengali' does not imply that she might not speak Arabic.

While I have formulated the examples with *both* and *either*, this is inessential. We observe the same contrast when we replace the second conjunct with 'you are not required to read *Beloved* and *Song of Solomon*' and 'you are permitted to skip *Beloved* or *Song of Solomon*', respectively.

Thus strong permission's reduction of free choice to SDA faces challenges of its own. The next section shows how to keep the reduction of free choice to SDA for weak permission, while avoiding the challenges facing strong permission. To do so, we appeal to *alternatives*.

5 Alternatives

A number of authors argue that conditional antecedents are interpreted not as a single proposition, but as a set of propositions, called the antecedent's 'alternatives', and that a conditional is true just in case, for each alternative, supposing it results in the truth of the consequent (Alonso-Ovalle 2006, 2009, Ciardelli 2016, Santorio 2018, Willer 2018, Cariani and Goldstein 2020, Khoo 2022).

5.1 Alternative semantics

Here I adopt Alonso-Ovalle's (2006) account, formulated within alternative semantics (Hamblin 1973, Kratzer and Shimoyama 2002).[7]

Our language is the same as the conditional reduction of deontic logic from section 2. In addition, we introduce a flattening operator (written '!', also known as 'existential closure' and written as '$\exists$'). Our models are the same as in Section 2.

In alternative semantics, each sentence denotes a set of propositions, where each proposition is in turn a set of worlds. Where A is a sentence and w a world, we let $[\![A]\!]^w$ denote A's semantic value at w according to alternative semantics.[8] A sentence is true at a world w just in case some alternative is true there: $w \in \bigcup [\![A]\!]^w$. To remove notational clutter, we let $|A|^w$ abbreviate $\bigcup [\![A]\!]^w$ and will often write $[\![A]\!]^w$ as $[\![A]\!]$ and $|A|^w$ as $|A|$. Entailment is defined as usual: A *entails* C (denoted $A \models C$) iff for every world where A is true, C is true.

The clauses for atomics and the connectives are:

(13) a. $[\![\mathrm{p}]\!] = \{V(\mathrm{p})\}$

[7]This choice is not essential. One might just as easily use other frameworks; notably, inquisitive semantics (Ciardelli 2016, Ciardelli, Groenendijk, and Roelofsen 2018). Note that alternative and inquisitive semantics assign different alternatives for Hurford disjunctions, such as $A \vee (A \wedge B)$, where one disjunct entails the other (Ciardelli and Roelofsen 2017). Hurford's constraint predicts these to be odd (Hurford 1974), unless the weaker disjunct is strengthened to an exhaustive reading, e.g. strengthening $A \vee (A \wedge B)$ to $(A \wedge \neg B) \vee (A \wedge B)$. Given that Hurford disjunctions are independently odd, we do not consider them below. Consequently, nothing in what follows hinges on the choice between alternative and inquisitive semantics.

[8]The interpretation function in alternative semantics is typically not defined to be relative to a world, while ours is. We make the change to impose definedness conditions on the interpretation function, since they will depend on the world of evaluation, as in (18a), (19a), (20a), and (21a) below. Apart from this difference, the change has no effect.

b. $[\![\neg A]\!] = \{\overline{\bigcup [\![A]\!]}\}$
c. $[\![A \wedge B]\!] = \{p \cap q : p \in [\![A]\!], q \in [\![B]\!]\}$
d. $[\![A \vee B]\!] = [\![A]\!] \cup [\![B]\!]$

Negations have a single alternative: the proposition that each of the negated sentence's alternatives is false. Conjunction denotes pairwise intersection, disjunction union.

The existential operator takes any sentence and returns one with a single alternative, leaving its truth conditions unchanged.

(14) $[\![!A]\!] = \{\bigcup [\![A]\!]\}$

To illustrate, $p \vee q$ has two alternatives, $[\![p \vee q]\!] = \{|p|, |q|\}$, while $!(p \vee q)$ has just one, $[\![!(p \vee q)]\!] = \{|p| \cup |q|\}$. (We will use this operator to account for counterexamples to SDA and the behaviour of modals and conditionals under probability operators, discussed in Section 6.)

Alonso-Ovalle (2006) proposes the following clauses for the conditional operators.[9]

(15) *Alonso-Ovalle's semantics for conditionals.*
a. $[\![A \mathrel{\Box\!\!\rightarrow} C]\!] = \{\{w' : \forall p \in [\![A]\!], f(p,w') \subseteq |C|\}\}$
b. $[\![A \mathrel{\Diamond\!\!\rightarrow} C]\!] = \{\{w' : \forall p \in [\![A]\!], f(p,w') \cap |C| \neq \emptyset\}\}$

These validate SDA for $\Box\!\!\rightarrow$ and $\Diamond\!\!\rightarrow$ alike.

(16) a. $(A \vee B) \mathrel{\Box\!\!\rightarrow} C \models (A \mathrel{\Box\!\!\rightarrow} C) \wedge (B \mathrel{\Box\!\!\rightarrow} C)$
b. $(A \vee B) \mathrel{\Diamond\!\!\rightarrow} C \models (A \mathrel{\Diamond\!\!\rightarrow} C) \wedge (B \mathrel{\Diamond\!\!\rightarrow} C)$

Some motivation for validating SDA for $\Diamond\!\!\rightarrow$ comes from Alonso-Ovalle (2006:18–19), who observes that *might* counterfactuals validate SDA. The point applies to possibility modals in general. 'If the die landed 1 or 2, you could win' has a prominent reading on which it implies both 'if the die landed 1, you could win' and 'if the die landed 2, you could win'.[10]

Conditionals in downward-entailing environments. Alonso-Ovalle's semantics predicts conditionals in downward-entailing environments to have a weak reading: true iff they are true for *some* alternative.[11] This predicts $\neg((A \vee B) \mathrel{\Box\!\!\rightarrow} C)$ to be equivalent to

[9] Cf. Simons' claim that 'when a modal (or other operator) takes a non-singleton set as argument, the operator in some sense interacts with each member of the argument set' (Simons 2005:278).

[10] Note that these data only motivate SDA for $\Diamond\!\!\rightarrow$ if we take 'might' and 'could' conditionals to express $A \mathrel{\Diamond\!\!\rightarrow} C$. This is controversial. Others have argued that they instead express $\Diamond(A \mathrel{\Box\!\!\rightarrow} C)$ or $A \mathrel{\Box\!\!\rightarrow} \Diamond C$. For discussion, see Stalnaker (1980), DeRose (1994), and Bennett (2003:189–93).

[11] A downward-entailing environment is one that reverses the direction of entailment, i.e. an operator O is downward-entailing iff whenever A entails B, $O(B)$ entails $O(A)$.

$\neg(A \boxrightarrow C) \vee \neg(B \boxrightarrow C)$.

Santorio (2018:524) observes that this is not quite right, offering the examples:

(17) a. I doubt that, if Alice or Bob went, the party would be fun.
b. No girl would be happy if she failed algebra or geography.

These show that conditionals in downward-entailing environments keep their strength: $\neg((A \vee B) \boxrightarrow C)$ is rather equivalent to $\neg(A \boxrightarrow C) \wedge \neg(B \boxrightarrow C)$. (17a) intuitively implies that the speaker doubts—i.e. thinks it is not the case—that if Alice went, the party would be fun, *and* doubts that if Bob went, the party would be fun. Likewise, (17b) implies that no girl would be happy if she failed algebra, *and* no girl would be happy if she failed geography.[12]

To account for these inferences, following Santorio (2018) and Cariani and Goldstein (2020), we add the following 'homogeneity' assumptions to the semantics of conditionals, to the effect that all of the antecedent's alternatives have the same status with respect to the consequent.[13] For a universal conditional, the assumption is that all or none of the antecedent's alternatives guarantee the consequent; for an existential conditional, that all or none of the antecedent's alternatives are compatible with the consequent.

(18) a. $[\![A \boxrightarrow C]\!]$ is defined only if
$\forall p, q \in [\![A]\!], f(p,w) \subseteq |C|$ iff $f(q,w) \subseteq |C|$;
b. if defined, $[\![A \boxrightarrow C]\!] = \{\{w' : \forall p \in [\![A]\!], f(p,w') \subseteq |C|\}\}$

(19) a. $[\![A \diamond\!\!\rightarrow C]\!]$ is defined only if
$\forall p, q \in [\![A]\!], f(p,w) \cap |C| \neq \emptyset$ iff $f(q,w) \cap |C| \neq \emptyset$;
b. if defined, $[\![A \diamond\!\!\rightarrow C]\!] = \{\{w' : \forall p \in [\![A]\!], f(p,w') \cap |C| \neq \emptyset\}\}$

5.2 Extending alternatives to deontic modals

The conditional reduction of deontic logic gives us an automatic recipe to take any semantics for the conditional and return a semantics for deontic modals. We take $\Box A$ to be

[12] Santorio's own example is 'None of my friends would have fun at the party if Alice or Bob went'. I have reformulated the example to avoid the objection that it is parsed with negation scoping only over the consequent, as [None of my friends would have fun at the party] [if Alice or Bob went], $(A \vee B) \boxrightarrow \neg C$.

[13] Alonso-Ovalle (2006:30–32) considers, but ultimately does not adopt, these homogeneity assumptions. His reason is that the following example (attributed to Angelika Kratzer) sounds consistent: 'It is plain false that Hitler would have been pleased if Spain had joined Germany or the U.S. ... There is enough evidence showing that he might have objected to Spain joining the U.S. If she had joined Germany, he would have been pleased, of course'. But as Santorio (2018:n. 25) notes, *it is plain false* that appears to also negate presuppositions: 'It is plain false that the King of France is bald, since there is no King of France' is also consistent. Given this, Kratzer's is compatible with the homogeneity assumptions after all.

equivalent to $\neg A \,\Box\!\!\rightarrow\, \neg\text{OK}$ and $\Diamond A$ to $A \,\Diamond\!\!\rightarrow\, \text{OK}$.[14] Paired with the above alternatives semantics of conditionals, this returns:

(20) a. $[\![\Box A]\!]$ is defined only if
$\forall p, q \in [\![\neg A]\!], f(p,w) \subseteq \overline{\text{OK}_w}$ iff $f(q,w) \subseteq \overline{\text{OK}_w}$;
b. if defined, $[\![\Box A]\!] = \{\{w' : \forall p \in [\![\neg A]\!], f(p,w') \subseteq \overline{\text{OK}_{w'}}\}\}$

(21) a. $[\![\Diamond A]\!]$ is defined only if
$\forall p, q \in [\![A]\!], f(p,w) \cap \text{OK}_w \neq \emptyset$ iff $f(q,w) \cap \text{OK}_w \neq \emptyset$;
b. if defined, $[\![\Diamond A]\!] = \{\{w' : \forall p \in [\![A]\!], f(p,w') \cap \text{OK}_{w'} \neq \emptyset\}\}$

This automatically guarantees free choice and its negative counterpart, called *Dual prohibition*.[15]

(22) a. *Free choice:* $\Diamond(A \vee B) \models \Diamond A \wedge \Diamond B$
b. *Dual prohibition:* $\neg\Diamond(A \vee B) \models \neg\Diamond A \wedge \neg\Diamond B$

This is the right result. 'Alice may not have an apple or a pear' intuitively implies both that she may not have an apple and that she may not have a pear. This improves on semantic accounts of free choice such as Zimmermann (2000), Geurts (2005), Simons (2005), and Aloni (2007), which derive a weak reading for $\neg\Diamond(A \vee B)$, predicting it to be equivalent to $\neg\Diamond A \vee \neg\Diamond B$. In addition, instead of merely stipulating the homogeneity inferences, as Goldstein (2019) does, we derive them from independently attested behaviour of conditionals.

In this way, we overcome the tension Prior identified, that permission is weak in some cases but strong in others, such as disjunction. The difference is one of alternatives. Permission is weak (quantifies existentially) within each alternative, but strong (quantifies universally) across alternatives.

For instance, this account correctly predicts the difference between (8), 'Alice may buy more than four tickets', and (9), 'Alice may buy five or more tickets'. The embedded sentences have the same truth conditions but different alternatives. 'Alice buys more than four tickets' is not disjunctive, and so has a single alternative. Plausibly, among the worlds we consider when supposing 'If Alice bought more than four tickets' is one where she buys exactly five. Since this world is OK, we predict (8) to be true. In contrast, 'Alice buys five or more tickets' is disjunctive, and so has multiple alternatives. For Alice to be allowed to buy five or more tickets, each alternative must be permitted. No world where she buys more than five is OK, so (9) is not true.

Lastly, we account for the contrast in (11) and (12) between $\neg\Box(A \wedge B)$ and $\Diamond(\neg A \vee \neg B)$. Since $\neg(A \wedge B)$ has a single alternative, $\neg\Box(A \wedge B)$ is true iff some selected $\neg(A \wedge$

[14] Here 'OK' denotes an atomic sentence whose valuation is the set of OK_w-worlds.

[15] For discussion of dual prohibition, see Fox (2007a), Romoli and Santorio (2019), Bar-Lev and Fox (2020), Marty and Romoli (2022), Tieu, Bill, and Romoli (2019), Goldstein (2019), and Aloni (2022).

B)-world is OK. We therefore predict (10a), $\Box A \wedge \neg\Box(A \wedge B)$, to be consistent. In comparison, $\Diamond(\neg A \vee \neg B)$ entails $\Diamond\neg A$, so (10b), $\Box A \wedge \Diamond(\neg A \vee \neg B)$, is contradictory.

Turning to obligation, the crucial observation is that, in alternative semantics, negations have a single alternative. This renders alternatives inert for obligation, making the resulting theory of obligation equivalent to the alternative-free theory from Section 2.1: $\Box A$ holds iff no selected $\neg A$-world is OK. As a result, 'Bobby must clean the kitchen or his bedroom' is true just in case if he does neither, the rules are broken. This seems right. In particular, we do not inadvertently predict $\Box(A \vee B)$ to entail $\Box A \wedge \Box B$, something we do not want to predict.

6 Ambiguity via flattening: counterexamples to SDA and probability

The theory developed so far is a semantic account of free choice: according to it, $\Diamond(A \vee B)$ *entails* $\Diamond A \wedge \Diamond B$. A natural worry that arises for semantic accounts concerns their predictions for embedded environments. A key test case comes from negation, which we have already handled, though the worry is more general. In this section, we see that a similar worry arises for possibility statements embedded under probability operators. While this initially seems to challenge a semantic account of free choice, I show that a solution is readily available within the conditional reduction of modality, again taking inspiration from existing work on conditionals.

Possibility statements exhibit interesting behaviour under probability operators. Consider:

> It's time for dessert. Alice's parents play a game to decide what dessert she is allowed to have. They roll a die. If it lands 1 or 2, she is allowed fruit (and only fruit). If it lands 3 or 4, she is allowed ice cream (and only ice cream). If it lands 5 or 6, she is allowed cake (and only cake). The die has just been cast, but we don't know how it landed. What is the probability that Alice is allowed to have ice cream or cake?

We can easily access a reading on which the answer is two thirds. After all, there is a two in three chance that the die lands 3, 4, 5 or 6.[16] At the same time, the probability that she is allowed ice cream *and* allowed cake is intuitively zero, since no matter how the die lands, she is allowed at most one dessert. If she is allowed one, she is not allowed the other.

These judgements challenge semantic theories of free choice. The problem is that entailment preserves probability: if A entails B, the probability of A cannot exceed that of B (Kolmogorov 1933). For instance, 'Alice is in Berlin' entails that she is in Germany,

[16] There is also intuitively a reading on which the answer is zero, which we come back to below.

so the probability that she is in Berlin cannot be higher than the probability that she is in Germany. Likewise, if $\Diamond(A \vee B)$ entails $\Diamond A \wedge \Diamond B$, the probability that the former is true cannot exceed the probability that the latter is true. In the dessert scenario, however, this is exactly what we found. The probability that she is allowed ice cream or cake was intuitively two thirds, while the probability that she is allowed ice cream and allowed cake was zero.

This appears to challenge a semantic account of free choice. However, as we will see, exactly the same issues have already arisen for disjunction in conditional antecedents and been addressed within the conditionals literature. The conditional reduction of modality allows us to automatically extend those accounts to solve this problem.

A number of authors argue that disjunctive antecedent conditionals are ambiguous between two readings, one that validates SDA and one that does not.[17] Within the current framework, we may implement an ambiguity approach using the flattening operator, taking the disjunction 'A or B' to be ambiguous between $A \vee B$ and $!(A \vee B)$.[18] The former validates SDA, while on the latter, its validity depends on the underlying selection function: it is valid iff $f(p,w) \cup f(q,w) \subseteq f(p \cup q,w)$.

I see two motivations for adopting an ambiguity approach.

6.1 Motivation 1: Counterexamples to SDA

It is well known that SDA has counterexamples, such as McKay and van Inwagen's (1977) classic:

(23) If WWII Spain had joined the Allies or the Axis, they would have joined the Axis.

If SDA were unrestrictedly valid, this would imply that if WWII Spain had joined the Allies, they would have joined the Axis, which is clearly false.

This suggests that SDA is not valid, but does not settle whether it is simply invalid, or ambiguous between an SDA-validating and -invalidating reading. Nute (1980a) offers the following argument for an ambiguity approach. Compare (23) with (24).

(24) If WWII Spain had joined the Allies or the Axis, Hitler would have been pleased.

[17] An ambiguity approach is endorsed by Hilpinen (1982), Loewer (1976), McKay and van Inwagen (1977), Warmbrōd (1981), Pollock (1984), Alonso-Ovalle (2009), Santorio (2018), and Khoo (2021b). Among those who claim SDA is valid are Nute (1975), Ellis, Jackson, and Pargetter (1977), Fine (2012), Starr (2014), and Willer (2018). Among those who claim it is invalid—at least on some readings—are Nute (1980b, 1984), Bennett (2003), van Rooij (2006), and Lassiter (2018).

[18] This is also explored by Cariani and Goldstein (2020).

There is a striking contrast between (23) and (24).[19] The first easily sounds true, while the second reads as false.[20]

An ambiguity approach can capture Nute's contrast by taking the conditionals' antecedents to have different readings: (23) expresses $!(A \vee B) \boxrightarrow B$, while (24) expresses $(A \vee B) \boxrightarrow C$. This captures the desired entailments: (23) does not entail that if WWII Spain had joined the Allies, they would have joined the Axis, while (24) does entail that if they had joined the Allies, Hitler would have been pleased—which explains why it sounds incorrect.

Without adopting an ambiguity approach, it is hard to see how to capture Nute's contrast. For then the two antecedents have the same meaning. It is widely assumed that two counterfactual antecedents with the same meaning, uttered in the same context, have the same counterfactual domain: the same set of worlds at which we evaluate the consequent. (Our formalisation above respects this: the counterfactual domain is determined by the contextually-supplied selection function, antecedent, and world of evaluation.) Given this, (23) and (24) have the same counterfactual domain. Then for (23) to come out true, Spain must join the Axis at every world in the domain. But at every such world, Hitler is pleased, so (24) comes out true too—the wrong result. An ambiguity approach gets around this problem by taking the two antecedents to express different propositions.

6.2 Motivation 2: Conditionals under probability operators

A second argument for the ambiguity approach comes from conditionals under probability operators. Consider the following case from Santorio (2021:633).

> Alice threw a six-sided die. Before her throw, Bob bet that the outcome would be a 3 or a 4. We don't know the outcome, but we hear two reports. Mary says that the die landed on 2, 3, or 4; Sue says that it landed on 3, 4, or 5. We have no idea whether the reports are accurate.

Consider:

[19] This casts doubt on explanations of SDA in terms of pragmatic shifts in context. Fine (2012:232), for example, proposes 'a principle of 'Suppositional Accommodation', according to which we always attempt to interpret a counterfactual in such a way that its antecedent *A* represents a genuine counterfactual possibility'. This rule only concerns whether the antecedent is counterfactually possible, and therefore does not predict changes in the *consequent* to affect SDA's validity, contrary to what we observe.

[20] That being said, the SDA-invalidating reading appears to be available given enough explicit contextual support. Nute (1980b:163) notes that the following sounds acceptable: 'if Spain had fought for the Allies or Axis, they would have fought for the Axis. So if Spain had fought for the Allies or Axis, Hitler would have been pleased'. The point is not decisive: one might reply that the sentence is interpreted enthymematically, as in, say, 'so if Spain had fought for the Allies or Axis, [they would have fought for the Axis, in which case] Hitler would have been pleased'. This fits the pattern above, where one of the alternatives entails the consequent.

(25) If Mary's report is accurate or Sue's report is accurate, probably Bob won.

Santorio observes that (25) has a true reading here.[21] Now, it is true on the SDA-validating reading, $(A \vee B)$ $\Box\!\!\rightarrow$ *probably* C, but false on the SDA-invalidating reading, $!(A \vee B)$ $\Box\!\!\rightarrow$ *probably* C. On the former reading, it is equivalent to 'if Mary's report is accurate, probably Bob won, and if Sue's report is accurate, probably Bob won', which is true.[22] On the latter reading, it is equivalent to 'If the die landed between 2 and 5, probably Bob won'. This is false since in that case he would only have a half chance of winning, while *probably* requires more than half chance.

Other examples bring out the SDA-invalidating reading. Consider:

(26) If the die landed 2, 3, or 4, probably Bob won.

This also has a true reading here. Now the pattern is reversed: $!(2 \vee 3 \vee 4)$ $\Box\!\!\rightarrow$ *probably* C is true but $(2 \vee 3 \vee 4)$ $\Box\!\!\rightarrow$ *probably* C is not, since it is false that if the die landed 2, probably Bob won—in that case he certainly lost. Thus to capture the behaviour of conditionals under probability operators, we need to appeal to both readings that the ambiguity approach to SDA provides.

Of course, an ambiguity approach is only predictive if we have a grasp on when each reading appears. The following empirical generalisation gives the right results for Nute's contrast and the probability data above.[23]

(27) *If A or B, C* may be parsed as $!(A \vee B)$ $\Box\!\!\rightarrow$ C when C entails $A \vee B$ or contains a probability operator.

It is an open and interesting question whether this empirical generalisation follows from more general principles.

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[21] As Santorio (2021) observes, this would be unexpected if SDA were an implicature—proposed by Bennett (2003), Klinedinst (2007), van Rooij (2010), Franke (2011), and Bar-Lev and Fox (2020)—since implicatures imply their literal meanings.

[22] We assume a standard semantics of *probably* here (see Yalcin 2010:928): *if A, probably C* is true just in case the selected A-worlds where C is true are more probable than the selected A-worlds where C is false: $Pr(f(A,w) \cap |C|)$ is greater than $Pr(f(A,w) \cap |\neg C|)$; equivalently, $Pr(|C||f(A,w)) > .5$, provided $f(A,w)$ is nonempty.

[23] This raises the question whether a McKay-van Inwagen-style counterexample could be constructed to flatten $\Diamond(A \vee B)$ to $\Diamond!(A \vee B)$, incorrectly predicting the disappearance of free choice in unembedded environments. However, while such examples are easy to construct for conditionals, they are impossible to construct for permission statements. To produce an analogue of McKay and van Inwagen's example for permission according to (27), we would need $\mathrm{OK}_w \subseteq |A|$ or $\mathrm{OK}_w \subseteq |B|$. Since the valuation of sentences and the interpretation of OK_w are separate components of the model, we can always find a model where these inclusions fail.

6.3 Permission under probability

The conditional reduction of modality allows us to apply this account of disjunction in conditional antecedents and to modals. In the dessert scenario, the 'two thirds' answer comes from the flattened reading of 'Alice is allowed to have ice cream or cake': $\Diamond!(A \vee B)$. In addition, there is intuitively a reading of the question on which the answer is zero or undefined.[24] This is the unflattened reading, $\Diamond(A \vee B)$, which entails $\Diamond A \wedge \Diamond B$. As discussed, in the dessert scenario, $\Diamond A \wedge \Diamond B$ is false, since no matter how the die lands, she will be allowed at most one dessert. An ambiguity approach—implemented via the flattening operator—correctly allows both readings.

7 Extensions of the theory

The conditional reduction serves as a general template to take any semantics of conditionals and return a theory of modals. This section shows how to extend this template in a variety of ways; in particular, (i) beyond deontic modals to arbitrary modal flavours, such as epistemic and ability modals; (ii) to a first-order language with quantifiers; and (iii) to account for deontic gaps, in which a statement's deontic status is left undecided (something Barker 2010 took to require a strong account of permission).

7.1 Different modal flavours

The inferences we are interested in also hold not only for deontic modals, but for other modal operators, such as epistemic and ability modals:

(28) a. Mr. X might be in Victoria or in Brixton. (Zimmermann 2000:ex. 6)

b. I can lick any man in the house, or drink the lot of you under the table. (Lewis 1977:ex. 4)

c. It is legal for you to report this as taxable income or for me to claim you as a dependent. (Lewis 1977:ex. 5)

In each case, $\Diamond(A \vee B)$ intuitively implies both $\Diamond A$ and $\Diamond B$, where $\Diamond$ represents epistemic, ability, and legal possibility, respectively.[25]

Here is a way to generalise the conditional reduction of modality to any modal flavour. We introduce modal operators $\Box_0, \Diamond_0, \Box_1, \Diamond_1, \ldots$, representing the different modal flavours, and assume that they have a standard semantics (Kripke 1959, Hanson 1965, Kratzer 1977): each modal flavour is associated with an accessibility relation

[24] In the dessert context, we predict $\Diamond(A \vee B)$ to suffer from presupposition failure, since its homogeneity presupposition fails. Thus whether its probability is zero or undefined depends on one's preferred theory of the interaction between presupposition failure and probability.

[25] Williamson (2007:157, 297) also considers analysing metaphysical possibility and necessity in terms of counterfactuals, along the same lines pursued in this section.

$R_0, R_1, \ldots$, where $R_i(w) = \{w' : wR_iw'\}$ denotes the set of R_i-accessible worlds from w. Necessity expresses universal quantification over the set of accessible worlds, possibility existential quantification. Let $[\![\cdot]\!]_s$ denote the interpretation function according to the standard theory.

(29) *The standard theory.*

a. $[\![\Box_i A]\!]_s = \{w' : R_i(w') \subseteq [\![A]\!]_s\}$

b. $[\![\Diamond_i A]\!]_s = \{w' : R_i(w') \cap [\![A]\!]_s \neq \emptyset\}$.

Let us now 'conditionalise' the standard theory, in the sense that we express it in terms of conditional selection functions and 'OK'. For each accessibility relation R_i, we introduce a corresponding conditional selection function f_i that takes a proposition p and a world w and returns the accessible worlds from w where p is true, and we define that a world is OK_i at w iff it is accessible from w.

$$f_i(p, w) = R_i(w) \cap p$$
$$\text{OK}_{i,w} = R_i(w)$$

We may even, if we like, plug in the standard theory of deontic logic, on which the deontically accessible worlds at a world are those that are deontically ideal from its standpoint (Hanson 1965). This differs from the implementation above, on which we took the selection function to be given by the semantics of counterfactuals. Which selection function to pick depends on whether one prefers the standard theory of deontic logic or the conditional reduction, a question I do not aim to settle here.

Given this choice of the selection function and 'OK', the standard theory and the conditional reduction collapse, in the following sense:

(30) a. $\{w' : R_i(w') \subseteq [\![A]\!]_s\} = \{w' : f_i([\![\neg A]\!]_s, w') \subseteq \overline{\text{OK}_{i,w'}}\}$

b. $\{w' : R_i(w') \cap [\![A]\!]_s \neq \emptyset\} = \{w' : f_i([\![A]\!]_s, w') \cap \text{OK}_{i,w'} \neq \emptyset\}$.

This equivalence only holds for the *alternative-free* conditional reduction. Unlike the standard theory, the conditional reduction easily integrates with alternatives: simply replace f with f_i in our modal entries above, (20) and (21). The resulting theory validates free choice for arbitrary modal flavours, thereby accounting for (28), while replicating the standard theory's predictions for $\Box A$ and $\Diamond A$ when A has a single alternative.

7.2 A first-order version

The parallel between SDA and free choice extends into the first-order domain. We see this with free-choice *any*. Vendler observes that if someone tells you to 'Take any one of them', they 'grant you the unrestricted liberty of individual choice' (Vendler 1967:79–80).

(31) a. You may take any card.
b. ⇒ For each card, you may take it.

Any in conditional antecedents behaves in a parallel fashion.

(32) a. If you took any card, the rules would be met.
b. ⇒ For each card, if you took it, the rules would be met.

I make the common assumption that *any* expresses existential quantification.[26] Then parallel to the inference from $\Diamond(A \vee B)$ to $\Diamond A \wedge \Diamond B$ is the inference from $\Diamond\exists xAx$ to $\forall x\Diamond Ax$, and parallel to the SDA inference from $(A \vee B) \;\Box\!\!\rightarrow C$ to $(A \;\Box\!\!\rightarrow C) \wedge (B \;\Box\!\!\rightarrow C)$ is the inference from $(\exists xAx) \;\Box\!\!\rightarrow C$ to $\forall x(Ax \;\Box\!\!\rightarrow C)$.

We can capture these inferences by providing a first-order version of alternative semantics. Our treatment here largely follows the first-order variant of inquisitive semantics given by Ciardelli, Groenendijk, and Roelofsen (2018:61). We assume a standard first-order language consisting of terms, function symbols, and relation symbols, where 0-place function symbols represent individual constants. To simplify the formalisation, we assume that for each element of the domain d, the language contains a constant d' whose interpretation is always d.

Our models have the form $M = (D, I, W, f, \mathrm{OK})$ where W, f, and OK are as before (Section 2), and the first-order specific components are D and I. $D = \{d_0, d_1, \dots\}$ is the domain, and I is an interpretation function taking each world w and returning a first-order structure (i.e. given a term, I_w returns an element of the domain, given an n-ary function symbol it returns a function $f : D^n \to D$, and given an n-ary predicate it returns a subset of D^n).

The clauses for the connectives, conditional and modal operators remain the same as in the propositional case. We replace the clause for the atomics and add the following clauses for the quantifiers:

(33) a. $[\![Ft_1, \dots, t_n]\!] = \{\{w' : (I_{w'}(t_1), \dots, I_{w'}(t_n)) \in I_{w'}(F)\}\}$
b. $[\![\forall xAx]\!] = \{p \cap q \cap \dots : p \in [\![Ad'_0]\!], q \in [\![Ad'_1]\!], \dots\}$[27]
c. $[\![\exists xAx]\!] = \bigcup_{d \in D} [\![Ad']\!]$

These are what we expect from treating universal quantification as generalised conjunction and existential quantification as generalised disjunction, where conjunction denotes pairwise intersection and disjunction union.

[26] For arguments that *any* is existential, see Horn (1972) and Ladusaw (1979). Kadmon and Landman (1993), Krifka (1995), Lahiri (1998), and Chierchia (2006) account for its behaviour assuming an existential semantics. Though this assumption is controversial: for an alternative view on which *any* is universal, see Carlson (1981) and Dayal (2004).

[27] More formally, this can be stated as $\{\bigcap_{d \in D} c(d) \mid c : D \to \wp(W)$, s.t. $c(d) \in [\![Ad']\!]$ for all $d \in D\}$, using a choice function c.

Pairing these clauses with the entries for $\Box\!\!\rightarrow$ and $\Diamond$ in (18) and (21) above, we predict $\Diamond\exists xAx$ to entail $\forall x\Diamond Ax$ and $(\exists xAx)\ \Box\!\!\rightarrow C$ to entail $\forall x(Ax\ \Box\!\!\rightarrow C)$, as desired.[28]

As in Section 6, we may also account for failures of these inferences using the flattening operator '!'.

(34) a. If anyone ate John's sandwich, he would be upset. $(\exists xFx)\ \Box\!\!\rightarrow Ga$

b. If anyone ate John's sandwich, Bob would. $(!\exists xFx)\ \Box\!\!\rightarrow Fb$

(34a), without the flattening operator, implies that for every salient person, if they ate John's sandwich, he would be upset, while (34b), with the flattening operator, does not.

Flattening furthermore allows us to predict the ambiguity of *any* under probability operators. Consider (35) in the context of the dessert scenario, where a die roll determines which one of three desserts Alice may eat:

(35) What is the probability that Alice may eat any of the desserts?

This has one reading on which the answer is '1' (no matter how the die lands, she will be allowed to eat some dessert or other) and another on which it is '0' (no matter how it lands, she will only be allowed to eat at most one dessert, and so will never have a choice of which dessert to eat). The first answer corresponds to reading the question as asking for the probability that $\Diamond!\exists xFx$ is true, the second for the probability that $\Diamond\exists xFx$ is true.

7.3 Deontic gaps

Barker (2010:12) argues that we need a strong account of permission to represent deontic gaps, which occur when something is neither permitted nor forbidden, pointing out that 'there is a difference between weak permission, which is the absence of prohibition, and strong permission [...] which is the assertion that some action is explicitly ok'.

If permission is strong, and we take a statement A to be forbidden ($F(A)$) just in case $A\ \Box\!\!\rightarrow \neg\text{OK}$ holds, then there is room for something to be neither permitted nor forbidden, since $A\ \Box\!\!\rightarrow \text{OK}$ and $A\ \Box\!\!\rightarrow \neg\text{OK}$ can simultaneously fail to be true. At first glance, weak permission cannot replicate this account of deontic gaps, according to which every

[28] It is worth mentioning that not every expression that can be expressed in terms of disjunction or existential quantification raises alternatives. In event semantics (Davidson 1967, Parsons 1990), for example, declarative statements express the existence of events. 'It snowed yesterday' says that a snowing event took place yesterday: $\exists e(\textit{actual}(e) \wedge \textit{snow}(e) \wedge \textit{runtime}(e) \subseteq \textit{yesterday})$. As Embry (2014) observes, this does not imply that if *any* snowing event whatsoever happened yesterday, I would have gone skiing (had it snowed 100 feet, I would have stayed home). Thus if we wish to represent the meaning of 'It snowed yesterday' using existential quantification over events, we should take these sentences to express the flattened quantifier $!\exists$ rather than the alternative-generating quantifier $\exists$. Having a meaning that we can express using disjunction or existential quantification is not enough for an item to generate alternatives. Its disjunctive/existential meaning must also be in some sense 'visible' to the alternative-generating mechanism.

statement is either permitted or forbidden: deontic gaps appear impossible.[29]

There is, however, an alternative account of deontic gaps, one that avoids this problematic prediction in mixed cases facing strong permission. It locates the source of deontic gaps not in the semantics of permission, but in the notion of 'OK'.[30] Deontic gaps intuitively occur when the relevant rules do not exhaustively classify all behaviour into two classes: those compatible with the rules and those incompatible with them. So far, however, we have assumed a simple, binary classification of worlds into 'OK' and 'not OK'. In cases of a deontic gap, our behaviour instead falls into one of *three* classes: those that meet the deontic ideals, those that violate them, and those whose deontic status is left undecided—the good, the bad, and the gappy.

Call a world 'OK' iff it is explicitly compatible with the rules, 'BAD' iff it is explicitly incompatible with them, and neither OK nor BAD when it is neither. Now weak permission can represent deontic gaps.[31]

(36) a. $P(A) \equiv A \diamond\!\!\rightarrow \text{OK}$
b. $O(A) \equiv \neg A \mathrel{\square\!\!\rightarrow} \text{BAD}$
c. $F(A) \equiv A \mathrel{\square\!\!\rightarrow} \text{BAD}$

These clauses allow deontic gaps. Whenever the set of selected A-worlds contains no OK-world and some world in the gap (that is neither OK nor BAD), A will be neither permitted nor forbidden. Crucially, however, permission is weak on this account. Deontic gaps and weak permission peacefully coexist. Given the viability of this alternative, then, the existence of deontic gaps does not motivate strong permission, contrary to

[29] Barker (2010:12) also states that 'Since I have claimed that *You may eat an apple or a pear* crucially neither permits nor forbids eating both an apple and a pear, we must use strong permission'. But cases where one sentence neither permits nor forbids another happen all the time; for instance, *You may dance* neither permits nor forbids singing. This does not show that we must use strong permission. What we have here is not a deontic gap but an *implication* gap, in which one sentence A implies neither another sentence B nor its negation—a perfectly ordinary occurrence. For there to be a deontic gap here, we must make the further step that B is neither permitted nor forbidden, which is a separate claim.

[30] One might think that deontic gaps are purely an epistemic matter, arising from our ignorance of the rules, in which case we would not need to modify the notion of OK. However, there is reason to think that they run deeper than that. If a justice system has never before ruled on whether a certain behaviour is legal, there is arguably no fact of the matter as to whether it is legal or illegal, not merely that it is in fact one or the other but we have not yet discovered which (see Kress 1989, Madry 1999).

[31] Note that this tripartite division goes beyond the Kratzerian analysis of OK from Section 2.1, which only divides worlds in two. It is an interesting open question how best to derive a tripartite division within a more general framework such as Kratzer's.

In addition, note that given such a tripartite division, we risk losing the equivalence of $\neg P(A)$, $O(\neg A)$, and $F(A)$. In contexts where we want to recover them, we may assume that the deontic ideals decide the status of the relevant worlds, categorising them as either OK or BAD, implemented via the following 'homogeneity' assumptions: $P(A)$ and $F(A)$ presuppose that every selected A-world is either OK or BAD, while $O(A)$ presupposes that every selected $\neg A$-world is either OK or BAD. With these assumptions, we recover the equivalence of $\neg P(A)$, $O(\neg A)$, and $F(A)$.

Barker (2010).

7.4 Double negation elimination

There is one area where alternative semantics' clause for negation may seem unusual: double negation. In alternative semantics, $\neg\neg(A \vee B)$ is not equivalent to $A \vee B$, since the first has one alternative, while the second has (at least) two. Paired with the present account, this makes a surprising prediction for sequences like (37).

(37) a. If Sue hired Alice or Bob, she would hire Alice. $!(A \vee B) \boxrightarrow A$
b. Sue must hire neither Alice nor Bob but she may hire Bob. $\Box\neg(A \vee B) \wedge \Diamond B$

Intuitively, this pair of sentences is inconsistent. However, as it stands, the present account predicts them to be consistent. According to it, $\Box\neg(A \vee B)$ is equivalent to $\neg\neg(A \vee B) \boxrightarrow \neg\text{OK}$. Since negations have a single alternative, this is equivalent to $!(A \vee B) \boxrightarrow \neg\text{OK}$. (37a) tells us that every selected $!(A \vee B)$-world is an A-world, so $!(A \vee B) \boxrightarrow \neg\text{OK}$ only tells us that A-worlds are not OK. It says nothing about the selected B-worlds, leaving open that some of them are OK ($\Diamond B$).

To get around the problem, one might explore a variant of alternative semantics that enforces the equivalence of $\neg\neg A$ and A, similar to what has been done in the case of anaphora (Gotham 2019, Hofmann 2022, Mandelkern 2022). The simplest—and admittedly most ad hoc—way to implement this is to keep the original clause from alternative semantics, $[\![\neg A]\!] = \{\overline{\bigcup [\![A]\!]}\}$, when A is not a negation and define $[\![\neg\neg A]\!] = [\![A]\!]$. This solves the problem, but it remains to be seen whether there are more elegant solutions available. Since our primary focus here is permission, we will not say more concerning this issue with obligation, though it is certainly worth future attention.

8 Comparisons

This section compares our account with existing accounts in the literature. We show how mixed cases pose a problem for all existing strong conditional analyses of permission (8.1) and how mixed cases differ from previous challenges to strong permission (8.2). We then compare our account with alternative theories that take negated conjunctions to have multiple alternatives (8.3).

8.1 Mixed cases in previous conditional analysis of permission

What do existing, strong conditional theories of permission predict for mixed cases? Answering this depends on the theory in question: although these theories have the same overall shape—holding that A is permitted iff a certain conditional 'if A, OK'

holds—they differ by interpreting the conditional in different ways. To my knowledge, six different interpretations of the conditional have been plugged in (Table 1).

Strong conditional theory	**Interpretation of the conditional 'if A, OK'**
Anderson (1967)	Relevant implication from relevance logic (Anderson and Belnap 1962)
von Wright (1968)	That A is a 'sufficient condition' for OK
Hilpinen (1982) and Nute (1985)	The variably strict conditional (Stalnaker 1968, Lewis 1973)
Asher and Bonevac (2005)	The defeasible conditional of non-monotonic logic (Asher and Morreau 1991, 1995, Morreau 1997)
Lokhorst (1997) and Barker (2010)	The conditional of linear logic (Girard 1987, Avron 1988)
Li (2025)	The dynamic strict conditional (von Fintel 2001b, Gillies 2007)

Table 1: Existing strong conditional theories of permission and their chosen interpretation of the conditional.

Mixed cases challenge all of them. We show this for each theory in turn.

Relevance logic. Anderson (1967) proposes a conditional reduction of deontic logic using the conditional of the relevance logic R. If we take permission to be strong, then the theory is that 'A is permitted' is true iff $A \to \text{OK}$ holds, where $\to$ is relevant implication. The trouble is that relevant implication in R is transitive, and therefore satisfies antecedent strengthening (with the strengthening understood in terms of relevant implication, i.e. A is stronger than B iff A relevantly implies B). Given that $A \to B$, from $B \to C$ we may infer $A \to C$. This analysis therefore predicts that whenever A relevantly implies B and B is permitted, A is permitted too: $A \to B, B \to \text{OK} \vdash_{\text{R}} A \to \text{OK}$.

Plausibly, 'the switches are both down' relevantly implies 'the switches are in the same position'. After all, the truth of the first guarantees the truth of the second, and they are clearly relevant to each other, being about the same subject matter. The strong conditional analysis, with relevant implication, therefore undesirably predicts 'the switches are allowed to be in the same position' to imply 'the switches are allowed to be both down'.

The strict and variably strict conditional. von Wright (1968) and Li (2025) propose a strong conditional analysis in terms of the strict conditional: 'A is permitted' expresses $\Box(A \supset \text{OK})$: every accessible world where A is true is deontically OK. Hilpinen (1982) and Nute (1985) suggest replacing the strict conditional with the variably strict conditional à la Stalnaker (1968) and Lewis (1973): 'A is permitted' expresses that the closest

worlds where A is true are OK.

These theories assume that natural language conditionals and permission statements make use of the same semantic components, and therefore predict permission statements to pattern with the corresponding natural language conditionals. As we saw in Section 3, they do not. In reply, one might consider keeping the strong analysis of permission, but disassociating it from the meaning of ordinary conditionals. In terms of selection functions, the thought is that conditionals use one selection function, permission statements another:

(38) *Disassociated strong permission.* There are two selection functions, f and f'.

a. $A \boxrightarrow C$ is true at w iff $f(|A|, w) \subseteq |C|$

b. $\Diamond A$ is true at w iff $f'(|A|, w) \subseteq \mathrm{OK}_w$

For this idea to work, the new selection function f' must be sensitive to the deontic ideals. To illustrate this, consider (1), where switch A must be up while B may be in any position. For the analysis in (38) to predict that the switches are allowed to be in the same position, the worlds selected by f' where they are in the same position must only contain worlds where they are both up. Now suppose the rules instead require A to be down. In that case, intuitively, the switches are still allowed to be in the same position. To predict this, now the selected worlds where they are in the same position must only contain worlds where they are both down. Changing the rules changes which worlds f' selects.

I see two ways to implement such sensitivity to the rules: one simple and one sophisticated. On the simple implementation, we take the selected worlds to be restricted to the OK worlds: $f'(p,w) = f(p,w) \cap \mathrm{OK}_w$. This, however, trivialises permission, predicting all permission statements to be true, since trivially, $f(p,w) \cap \mathrm{OK}_w \subseteq \mathrm{OK}_w$.

On the sophisticated implementation, the selection function is not restricted to the OK worlds, but merely prioritises them. We may implement this on a variably strict semantics like Lewis's using a similarity order. The idea is that conditionals are sensitive to the usual similarity order $<_w$, where $f(p,w)$ returns the closest worlds to w where p is true, while permission statements are sensitive to a modified order $<'_w$ that prioritises the OK worlds: every OK-world is promoted to be closer than any not OK-world, while within each class, the old similarity order prevails.[32] However, this sacrifices the strong theory's desired reduction of free choice to SDA. Consider a world where taking an apple is permitted but taking a pear is forbidden. Then the new order will rank worlds where you take an apple closer than worlds where you take a pear, and therefore predict 'You may take an apple or a pear' to be true—the wrong result.

[32] Formally: $u <'_w v$ iff either u is OK but v is not, or else u and v have the same deontic status—u is OK iff v is—and $u <_w v$. Assuming the limit assumption, we take $f(p,w) = \{w' \in p : \neg\exists w'' \in p, w'' <_w w'\}$ and $f'(p,w) = \{w' \in p : \neg\exists w'' \in p, w'' <'_w w'\}$.

Non-monotonic logic. Asher and Bonevac (2005) propose a strong conditional analysis of permission using the conditional from non-monotonic logic. They interpret the selection function $f(A,w)$ as returning 'the most normal A-worlds relative to w'. They gloss the conditional as follows: 'If A, then normally B is true at world w iff B is true in all normal A-worlds, that is, all A-worlds in which conditions are suitable for assessing (relative to w) what happens when A' (2005:309).

Normality, however, does not seem to be at play in the examples in (1)–(4). Take (3). Alice and Bob are intuitively permitted to talk to each other at work, but there is no suggestion whatsoever that doing so in French is more 'normal' than in English—quite the opposite, in fact. The way they actually do so, and have previously done so, is to talk in English, which is forbidden. The 'normal' worlds in this case are most plausibly those where they continue their old habit of communicating in English. Instead of delivering the intuitive judgement that Alice and Bob are permitted to talk to each other at work in this context, Asher and Bonevac predict that Alice and Bob are *forbidden* from doing so.

Linear logic. Lokhorst (1997) and Barker (2010) propose a conditional analysis of permission within linear logic. They propose that 'A is permitted' is equivalent to $A \multimap \mathrm{OK}$, where $\multimap$ is linear implication.[33] Linear implication is a kind of material implication: $A \multimap B$ is equivalent to $A^{\perp} \parr B$ and to $(A \otimes B^{\perp})^{\perp}$, where $^{\perp}$ is linear negation, $\parr$ is multiplicative disjunction, and $\otimes$ is multiplicative conjunction (these are one of the two kinds of disjunction and conjunction, respectively, provided by linear logic, the other being additive).

For this reason, linear implication inherits some of the properties of material implication familiar from classical logic. As Barker (2010:13) observes, in a statement permitting A, A occurs in a downward-entailing position (which, as Barker notes, is crucial in deriving the desired entailments). This has undesirable consequences for our mixed cases, which show that permission is not downward-entailing. 'The switches are both down' entails 'The switches are in the same position', but as we saw in (1), 'The switches are allowed to be in the same position' does not entail 'The switches are allowed to be both down'. Similarly, 'Alice and Bob talk to each other at work in English' entails 'Alice and Bob talk to each other at work', but in the context of (3), 'Alice and Bob are permitted to talk to each other at work' does not entail 'Alice and Bob are permitted to talk to each other at work in English'.[34]

[33] Lokhorst and Barker use 'δ' for 'OK', but its intuitive interpretation is the same.

[34] To show the problem more formally, we can construct a model of the mixed cases within linear logic. We will use the algebraic semantics due to Avron (1988), one also used by Barker (2010:27–29). Take the switches case. The most obvious model of this scenario within linear logic features four basic states: both switches up ($\uparrow\uparrow$), A up and B down ($\uparrow\downarrow$), A down and B up ($\downarrow\uparrow$), and both down ($\downarrow\downarrow$). The algebra is given by $\mathscr{A} = \wp(\{\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow\})$, where $\wedge = \circ = \cap$, $\vee = \cup$, the bottom element is $\emptyset$, the top element

In summary, all existing strong conditional theories of permission make incorrect predictions for mixed cases.

8.2 Comparison with previous criticisms of strong permission

It is worth distinguishing the challenge that mixed cases pose to strong permission from one often made, that something can be permitted even if not every 'way' for it to obtain is permitted; in other words, that permission is *coarse*.[35] Recall Prior's observation above, that 'α (say my feeding this man) is usually considered permissible even if some ways of doing α (e.g. feeding him with poison) are not, so long as some are' (Prior 1959:178). Kamp (1973:63) makes a similar point:

> What, for example, are the actions rendered permissible by my statement ['Michael may go to the beach']? Surely not every way of making the statement 'Michael goes to the beach' true. He is not to go by taxi; he is not to go to that section of the beach where there is no swimming guard. He is not to put on the coat which his mother bought him last month and which he is only to wear on special occasions and with explicit permission. And of course, as always he is to keep out of mischief and not to engage in behaviour that is offensive to others. All these ways of going to the beach remain proscribed.

The strong conditional theory has a plausible response to these objections. The conditional selection function is itself coarse: when we interpret a conditional, we do not consider *every* way for its antecedent to be true (Embry 2014). The strong conditional theory predicts permission to be accordingly coarse. When we suppose 'if I feed this man' we typically do not consider any worlds where one feeds him with poison. The conditional 'If I feed this man, I will not feed him with poison' is intuitively true, at least in ordinary contexts. The same applies to Kamp's example. When we suppose that Michael goes to the beach, we do not consider every way for him to do so. We might well accept a conditional such as 'If Michael goes to the beach, he will not go by taxi, and will swim where there is a swimming guard'.

The difference between these cases and the mixed cases from Section 3 is that, in these cases, the permission statement 'A is permitted' and the corresponding conditional

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$t = \{\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow\}$, and negation is set complementation: $\sim x = t \setminus x$. The rules are met just in case switch A is up: 'OK' is true at $\{\uparrow\uparrow, \uparrow\downarrow\}$. 'The switches are in the same position' is true at $\{\uparrow\uparrow, \downarrow\downarrow\}$, and 'the switches are both down' at $\{\downarrow\downarrow\}$. (We may represent these in linear logic as $(A \otimes B) \oplus (A^{\perp} \otimes B^{\perp})$ and $A^{\perp} \otimes B^{\perp}$, respectively, where $v(A) = \{\uparrow\uparrow, \uparrow\downarrow\}$ and $v(B) = \{\uparrow\uparrow, \downarrow\uparrow\}$.) Lokhorst and Barker then predict 'The switches are allowed to be in the same position' and 'the switches are allowed to both be down' to be equivalent in this model (specifically, they are both true at $\{\uparrow\uparrow, \uparrow\downarrow, \downarrow\downarrow\}$). This is clearly an undesirable result.

[35] The term 'coarseness' comes from Cariani (2013), who argues that *ought* is coarse; for discussion, see Bronfman and Dowell (2018) and Van De Putte (2019).

'if A, OK' pattern in the same way, whereas in the mixed cases, there is a clear contrast between the two.

8.3 Alternatives from negation

We have assumed that negations have a single alternative (double negations aside, discussed in Section 7.4). In contrast, Schulz (2018) and Romoli, Santorio, and Wittenberg (2022) propose that they have multiple alternatives. For example, both approaches represent a negated conjunction $\neg(A \wedge B)$ as having three alternatives: $A \wedge \neg B$, $\neg A \wedge B$, and $\neg A \wedge \neg B$. Paired with the present theory, this assigns $\neg\Box(A \wedge B)$ a remarkably strong meaning, predicting it to entail $\Diamond(A \wedge \neg B)$, $\Diamond(\neg A \wedge B)$, and $\Diamond(\neg A \wedge \neg B)$.[36]

This seems too strong. Consider:

(39) You don't have to read both *Beloved* and *Song of Solomon*, but you do have to read at least one of them. $\neg\Box(A \wedge B) \wedge \Box(A \vee B)$

This is perfectly consistent. But if 'you don't have to read both', $\neg\Box(A \wedge B)$, implied 'you are permitted to read neither', $\Diamond(\neg A \wedge \neg B)$, we would expect (39) to be as bad as 'You may have ice cream or cake, but you may not have cake', which it is not. After all, their unacceptability would be generated via the same mechanism: inferences from alternatives.

Similarly, recall from (10a) that $\Box A \wedge \neg\Box(A \wedge B)$ is consistent. In this context, where reading *Beloved* is obligatory, $\neg\Box(A \wedge B)$ is true but $\Diamond(\neg A \wedge B)$ is false.

Epistemic modals behave similarly. The following are consistent.

(40) a. Alice might not have read both books, but she read at least one of them.
b. Out of *Beloved* and *Song of Solomon*, Alice must have read *Beloved*, but she might not have read both.

Certainly, they are far better than 'Alice might have read *Beloved* or *Song of Solomon*, but she didn't read *Beloved*'.

In contrast, if we follow alternative semantics in assuming that negated conjunctions have a single alternative (the negated conjunction itself), we predict $\neg\Box(A \wedge B)$ to be true if supposing $\neg(A \wedge B)$ returns at least one OK world. In general, this could be a world where only A is true, a world where only B is true, or a world where both are false (provided there are such worlds among the selected $\neg(A \wedge B)$-worlds): we merely require the existence of some such world. This correctly predicts $\neg\Box(A \wedge B)$ to be in general compatible with $\Box(A \vee B)$, $\Box A$, and $\Box(A \wedge \neg B)$.

[36] This is because $\neg\Box(A \wedge B)$ presupposes that $(A \wedge \neg B) \,\Box\!\!\rightarrow \neg$OK, $(B \wedge \neg A) \,\Box\!\!\rightarrow \neg$OK, and $(\neg A \wedge \neg B) \,\Box\!\!\rightarrow \neg$OK have the same truth value, and hence that $(A \wedge \neg B) \,\Diamond\!\!\rightarrow$ OK, $(B \wedge \neg A) \,\Diamond\!\!\rightarrow$ OK, and $(\neg A \wedge \neg B) \,\Diamond\!\!\rightarrow$ OK do too. It also implies that one of these is true, and hence that all of them are.

9 Conclusion

A wide variety of authors have been attracted by the thought that there is a systematic connection between modals and conditionals: that the semantics of each draws from the same source. So far, this view has been construed as saying that '*A* is permitted' has a meaning roughly akin to: if *A* were true, the relevant deontic ideals would be met. One compelling motivation for this view comes from its account of free choice permission. But existing theories within this tradition fail to account for mixed cases. These show that for *A* to be permitted, it is enough that supposing *A* return *some* world where the deontic ideals are met. In addition, the strong conditional analysis of free choice permission faces counterexamples of its own.

To overcome these problems, we developed a novel account of permission on which permission is weak. While this runs directly into the problem of free choice permission, we showed how to overcome it by taking a leaf from the conditionals literature; in particular, using alternatives. The resulting theory simultaneously accounts for mixed cases and validates free choice. Borrowing from the conditionals literature again allowed us to make improved predictions for permission statements under probability operators. We then showed how to generalise the account to arbitrary modal flavours and to *any*.

Let us end by considering one exciting area for future work. There has been much interest in wide scope free choice, the inference from $\Diamond A \vee \Diamond B$ to $\Diamond A \wedge \Diamond B$ (Zimmermann 2000, Geurts 2005, Schulz 2005, Aloni 2022). Zimmermann illustrates this with the following examples:

(41) You may go to Brixton or you may go to Victoria.
   a. $\rightsquigarrow$ You may go to Brixton
   b. $\rightsquigarrow$ You may go to Victoria.

(42) He might be in Brixton or he might be in Victoria.
   a. $\rightsquigarrow$ He might be in Brixton.
   b. $\rightsquigarrow$ He might be in Victoria.

In light of the conditional reduction of modality, one would expect conditionals to behave in the same way, that 'if *A*, *C* or if *B*, *C*' intuitively implies 'if *A*, *C*' and 'if *B*, *C*'. Quite remarkably, this prediction is borne out (Wiggins and Edgington 1997, Geurts 2005, Khoo 2021a, McHugh 2025).

(43) If you had gone to Brixton, you would have been fine, or if you had gone to Victoria, you would have been fine.
   a. $\rightsquigarrow$ If you had gone to Brixton, you would have been fine.
   b. $\rightsquigarrow$ If you had gone to Victoria, you would have been fine.

This suggests that the parallel between modals and conditionals may run deeper than

is captured by the current theory, which does not capture wide scope free choice. It remains to be seen whether an extension of the present account can capture wide scope free choice and the conjunctive reading of disjunctions of conditionals in a uniform way.

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